Formula 1 Podium Prediction

Milestone 1 Analytical Report (companion to the notebook *Team 52 F1 Podium Prediction.ipynb*)

Team 52

58-7882	Ali Amr Hassan Elharras
58-3014	Ahmed Hany Ibrahim Elhussieny Mashaal
58-1804	Marwan Yasser Fahmy Gaber Mohamed
58-1753	Ahmed Mohamed Sayed Ahmed Abdelghaffar

Summary

We predict, for every driver in every race, whether they finish in the top 3, using only what is known after qualifying and before the race starts. The target is `podium = 1` when `positionOrder <= 3`, and each row is one driver in one race.

- **Data:** Kaggle/Ergast Formula 1 dataset, 14 tables, 1950-2024 (1,125 races, 26,759 result rows).
- **Modelling table:** 24,571 rows (one per driver per race), 15 features in 6 groups, podium rate 13.7%.
- **Final model:** logistic regression, chosen on validation PR-AUC (0.739). On the 2022-2024 test set it gets ROC-AUC 0.936, PR-AUC 0.708 and F1 0.685. Gradient boosting had a slightly higher test PR-AUC (0.721), but it was worse on validation.
- **Main findings:** grid position matters more than anything else. Front row conversion went up after 2014 and the circuit ranking changed. There is no home advantage once grid position is taken into account. Mechanical retirements dropped from 9.5% to 5.8% of starts.

1. Data audit and EDA before cleaning

We loaded all 14 tables as they are, without converting anything to missing, so we could count the placeholders first. All checks are done in code.

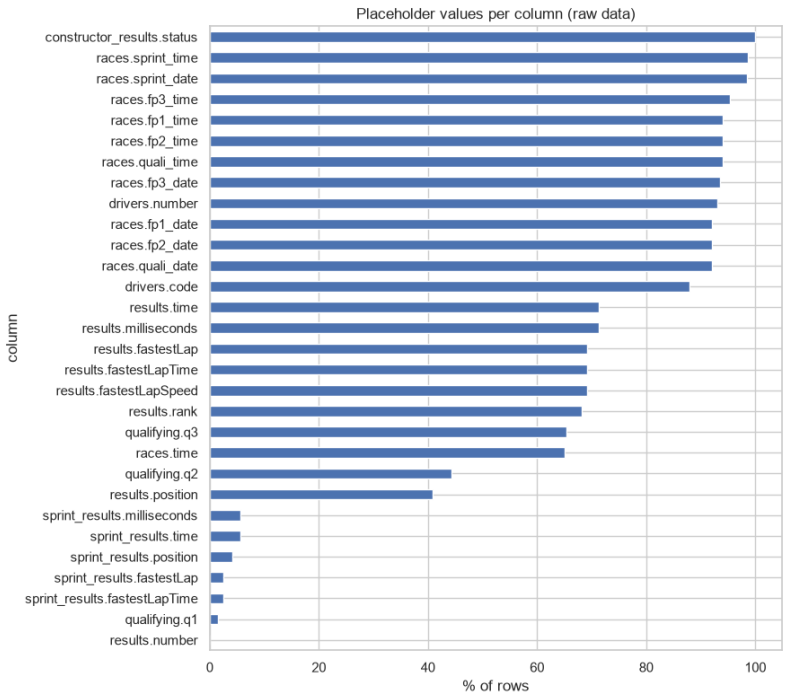


Figure 1. Share of rows that are \N or empty, per column (raw data).

- Most of the empty columns are practice session times or post race data (fastest lap, race time), which we don't use as features anyway.

- `results.position` is empty for about 41% of rows because those drivers were not classified. That's why the target uses `positionOrder`, which is always filled.
- q2 and q3 are empty for drivers knocked out earlier in qualifying, and the qualifying table also has 68 empty cells as a second kind of placeholder.

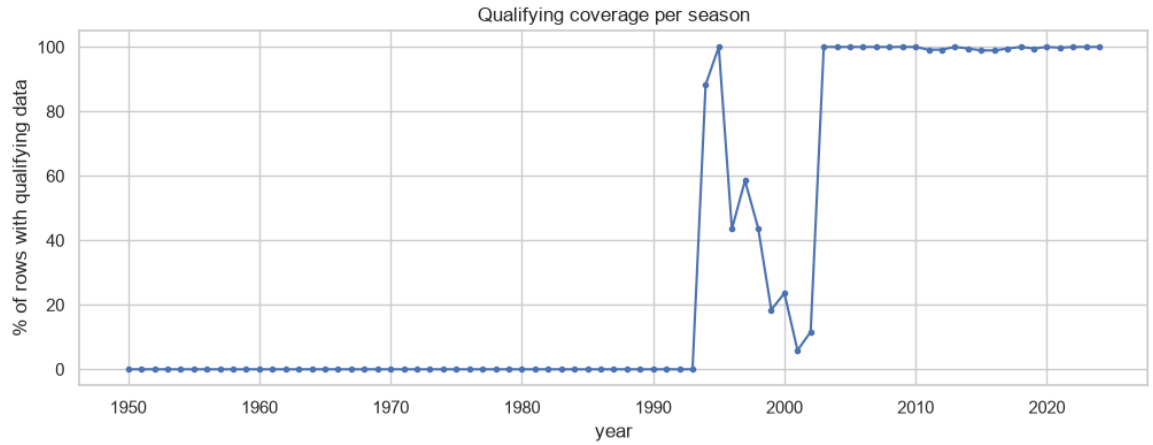


Figure 2. Qualifying coverage per season.

Qualifying coverage drift: there is no qualifying data before 1994, only 6-58% in 1996-2002, and full coverage from 2003. This affects how we handle the qualifying features (Section 5) and which years we train on (Section 6).

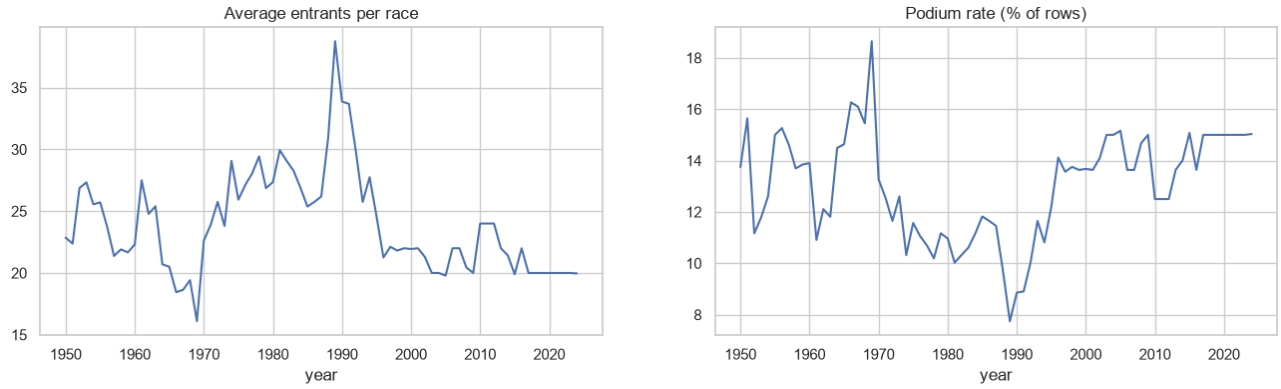


Figure 3. Average entrants per race and podium rate per season (raw data).

- Fields peaked around 1989 with about 39 entrants per race (non qualifiers still get a row). Today it is about 20.
- There are always 3 podium places, so the podium rate goes down when fields get bigger (about 8% in 1989, about 15% now). Overall it is 12.7%, so the classes are imbalanced.
- 3,397 rows have `positionOrder <= 3` but there are only 3,375 podium places (1,125 x 3). The extra rows come from shared drives in the early years.
- We also found 176 rows in duplicate driver/race pairs and 1,638 rows with grid = 0.

2. Cleaning

All cleaning is done on copies, the raw CSV files are never changed.

Step	What we did	Result
Placeholders	Read the files again with \N and empty strings treated as missing.	Every placeholder became NaN (e.g. 122,887 in results) and 10 text columns became numeric.
Dates	`races.date` and `drivers.dob` converted to dates.	All 1,125 races have a valid date.
Lap times	Strings like "1:27.452" converted to seconds (Q1, Q2, Q3, fastest lap).	Checked against `lap_times.milliseconds`: maximum difference 0.
Impossible times	Qualifying times slower than 150% of the fastest time in the same session set to missing.	2 values removed (1995 Japanese GP Q1, 2023 Austrian GP Q2).
Primary keys	Checked unique and non-null in all 14 tables.	All 14 pass.
Foreign keys	Checked that every ID exists in the table it points to (23 links).	No orphans.

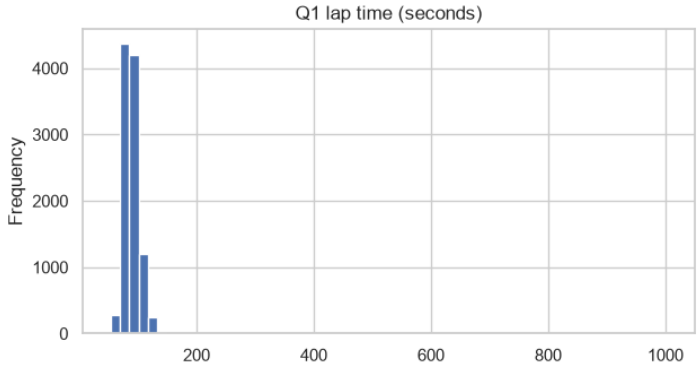


Figure 4. Q1 times in seconds before the outlier rule. The axis going up to about 1,000 s showed us the impossible value.

3. Grain: one row per driver per race

Each row is one prediction, so (raceId, driverId) has to be unique. We made three decisions about which rows to keep.

Decision	Rule	Why
Indianapolis 500 (1950-1960)	Removed (11 races, 405 rows).	Only 4 of its 107 drivers ever raced in another championship race, so it was basically a separate series. It would also distort the home race question.
Non-starters	Removed rows with F (failed to qualify), W (withdrawn), or grid = 0 with 0 laps. 1,719 rows.	They never raced, so there is nothing to predict, and none of them had a podium.
grid = 0	Kept as pit lane starts.	The 72 remaining rows all completed laps and are all from 2015-2024.
Shared drives	Kept the row with the best positionOrder (64 rows removed).	In all 61 pairs both rows have a non-zero grid, so we can't tell which car the driver started in. The best position matches the official result (e.g. Moss, 1957 British GP).

Step	Rows
All result rows	26,759
Without Indianapolis 500	26,354
Without non-starters	24,635
One row per driver per race (checked with an assert)	24,571

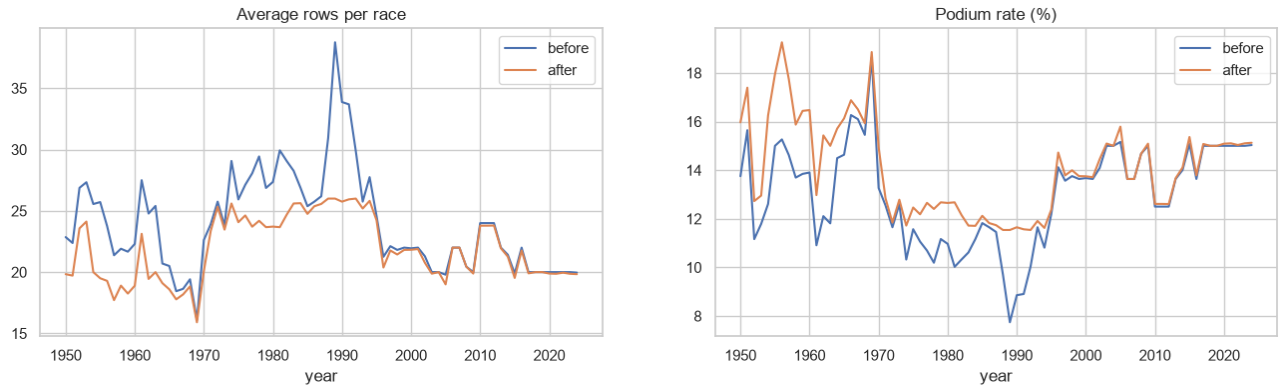


Figure 5. EDA after cleaning: rows per race and podium rate, before vs after.

After cleaning, the 1989 peak drops from about 39 to about 26 rows per race and the podium rate goes up in those crowded years, while modern seasons stay the same. The overall podium rate goes from 12.7% to 13.7%.

4. Data-engineering questions

All three questions use the cleaned table (one row per driver per race, starters only, no Indy 500) joined with other tables.

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

We joined results with qualifying, races and circuits. Front row means position 1 or 2, and we measured it both on the grid (after penalties) and in qualifying. We compare 2003-2013 with 2014-2024 because qualifying data is only complete from 2003. Conversion is podiums divided by front row starts, and we only rank circuits with at least 5 races in an era.

Era	Grid conversion	Qualifying conversion
2003-2013	68.3%	67.6%
2014-2024	76.2%	74.2%

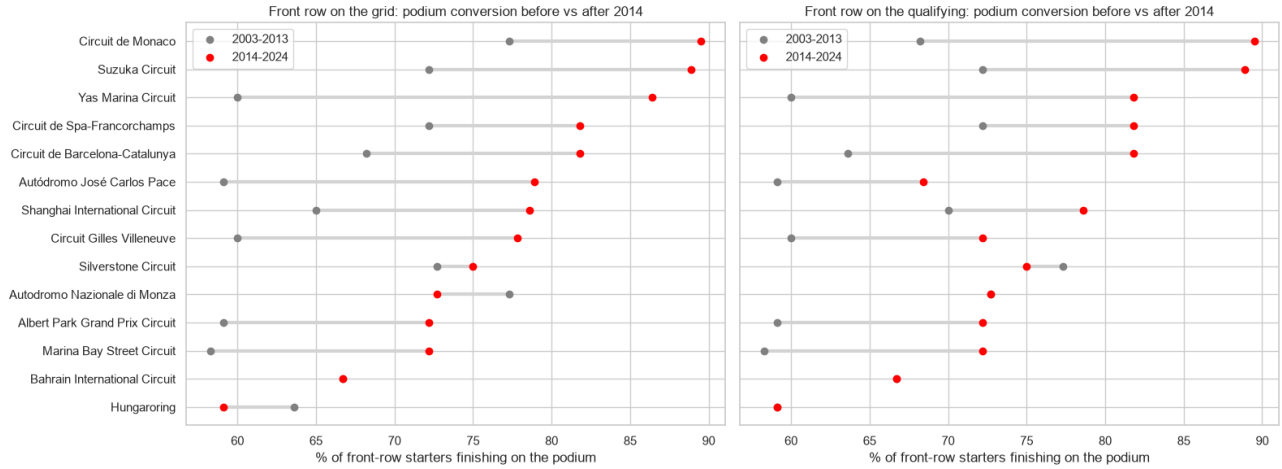


Figure 6. Front row conversion by circuit before (grey) and after (red) 2014, using grid (left) and qualifying (right).

Answer.

- Best circuits: Magny-Cours (92%), Istanbul Park (79%), Monza and Monaco (77%) in 2003-2013, and Monaco (90%), Suzuka (89%) and Yas Marina (86%) in 2014-2024.
- Yes, the ranking changed. Conversion went up overall (68% to 76%) but not evenly: Yas Marina (+26 points), Interlagos and Montreal went up the most, and Monza and the Hungaroring were the only ones that dropped. The rank correlation between the two eras is only 0.36 for grid and 0.54 for qualifying.

- Grid and qualifying front rows differ for 36 drivers in each direction because of penalties, but the circuit ranking is almost the same (rank correlation 0.92).
- Grain is driver/race rows from 2003-2024. 21 rows have no qualifying record, and 5 front row qualifiers who never started are not counted (e.g. Leclerc, Monaco 2021). Grid and qualifying are both known before the race. Each circuit only has 12-22 front row starts per era, so small differences are not reliable.

Q2. Do drivers podium more often at home, after controlling for starting position?

We joined results with drivers (nationality) and with races and circuits (country), using a nationality to country mapping we made by hand. To control for starting position we compare home and away drivers in the same grid band, and we also fit a logistic regression with home and grid band (`podium ~ home + grid band`). Pit lane starts are left out.

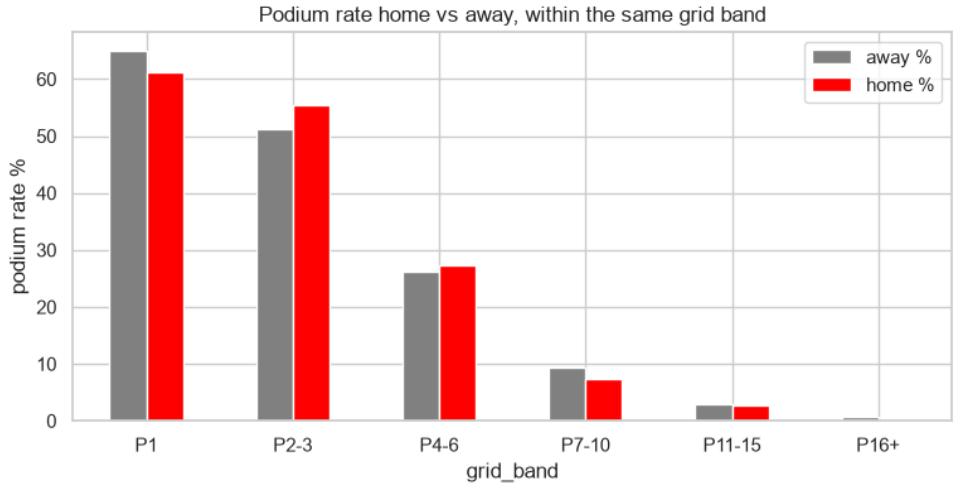


Figure 7. Podium rate at home vs away, within the same grid band.

Answer: no home advantage.

- Without controlling for grid, home drivers actually podium less (11.9% vs 13.9%), mostly because many home entrants were local drivers starting near the back.
- Inside each grid band the differences are small and go both ways. The regression gives an odds ratio of 0.96 (95% CI 0.81-1.14, p = 0.66), so there is no evidence of a home effect.
- 24,499 driver/race rows (1,928 at home). Every nationality is mapped, and both home status and grid are known before the race.

Q3. Which constructors have the highest mechanical-retirement rate in 2014-2021 vs 2022-2024?

We joined results with status, constructors and races, and put every status into finished, mechanical, accident/driver or other. Some calls we had to make: a tyre failure counts as mechanical but a puncture counts as an accident, a broken front wing counts as an accident, and "Retired" with no reason goes to other. The rate is mechanical retirements divided by starts, and a team needs at least 40 starts in an era.

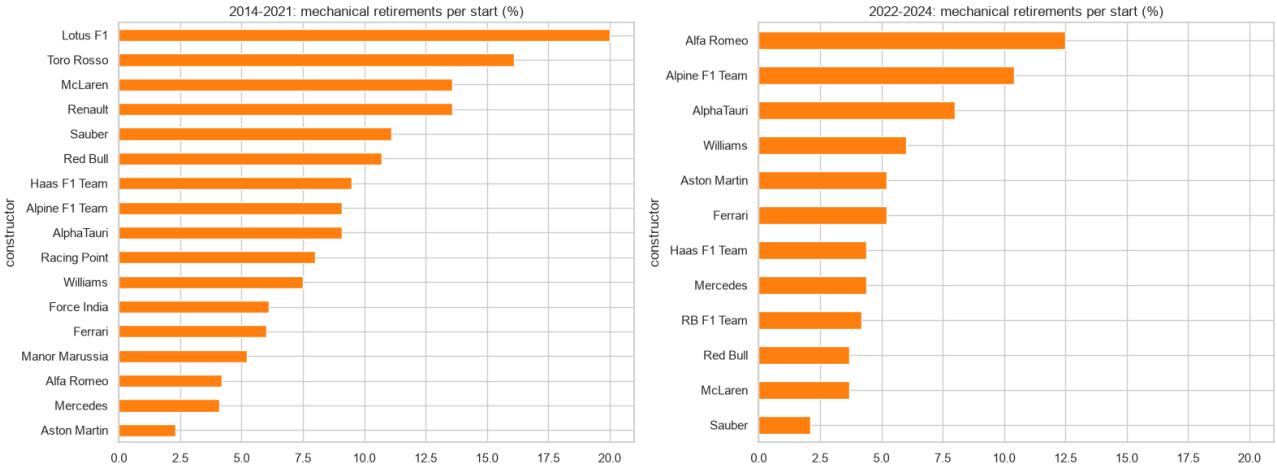


Figure 8. Mechanical retirements per start (%), by constructor and era.

Answer.

- 2014-2021: the highest were Lotus F1 (20.0%, only 2014-2015), Toro Rosso (16.1%), McLaren (13.6%) and Renault (13.6%). Mercedes was the lowest (4.1%).
- 2022-2024: the highest were Alfa Romeo (12.5%), Alpine (10.4%) and AlphaTauri (8.0%). Sauber (2.1%), McLaren and Red Bull (3.7%) were the lowest.
- Overall reliability improved from 9.5% to 5.8% of starts. McLaren went from one of the worst to one of the best, and Alfa Romeo went the other way.
- Status is only known after the race. That's fine for this question, but we never use it as a feature. With about 135 starts per team in 2022-2024 the exact order isn't very stable.

5. Feature engineering and selection

Every feature has to be known before the race starts. Form features only use previous races (we sort by date and shift every rolling window by one race), and since the standings tables are stored after each race, we use the standings after the previous race. To check this we rebuilt all features using only races before 2015, and all 20,378 pre-2015 rows came out the same, so nothing uses future data. We didn't use the sprint results because they only cover about 18 races and in 2021-2022 they already decided the grid.

We chose the features using the training seasons only (up to 2019), so the test years had no influence.

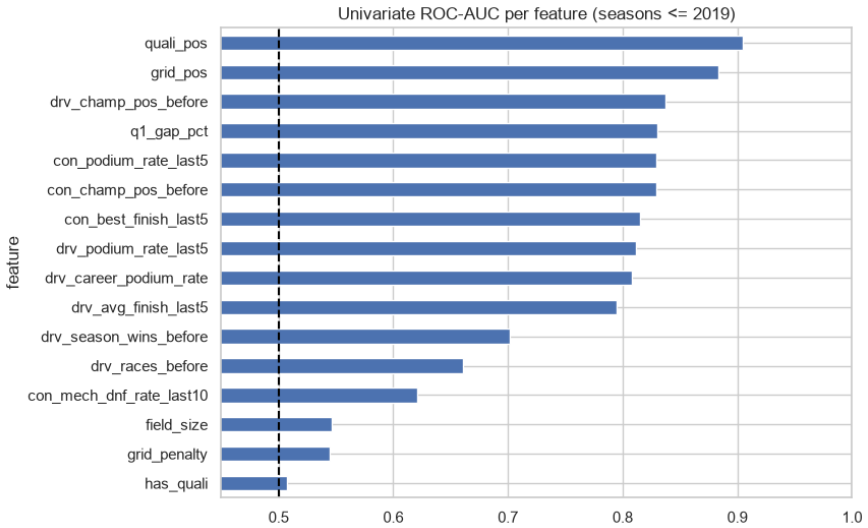


Figure 9. Univariate ROC-AUC of each candidate feature (seasons up to 2019).

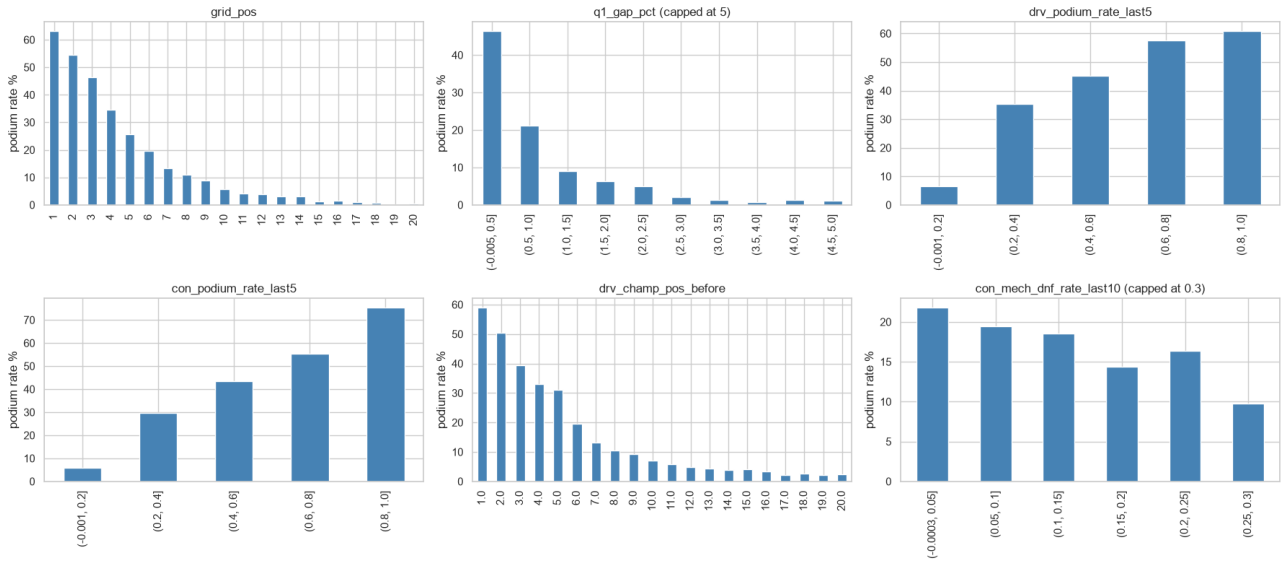


Figure 10. Podium rate across the range of six key features (seasons up to 2019).

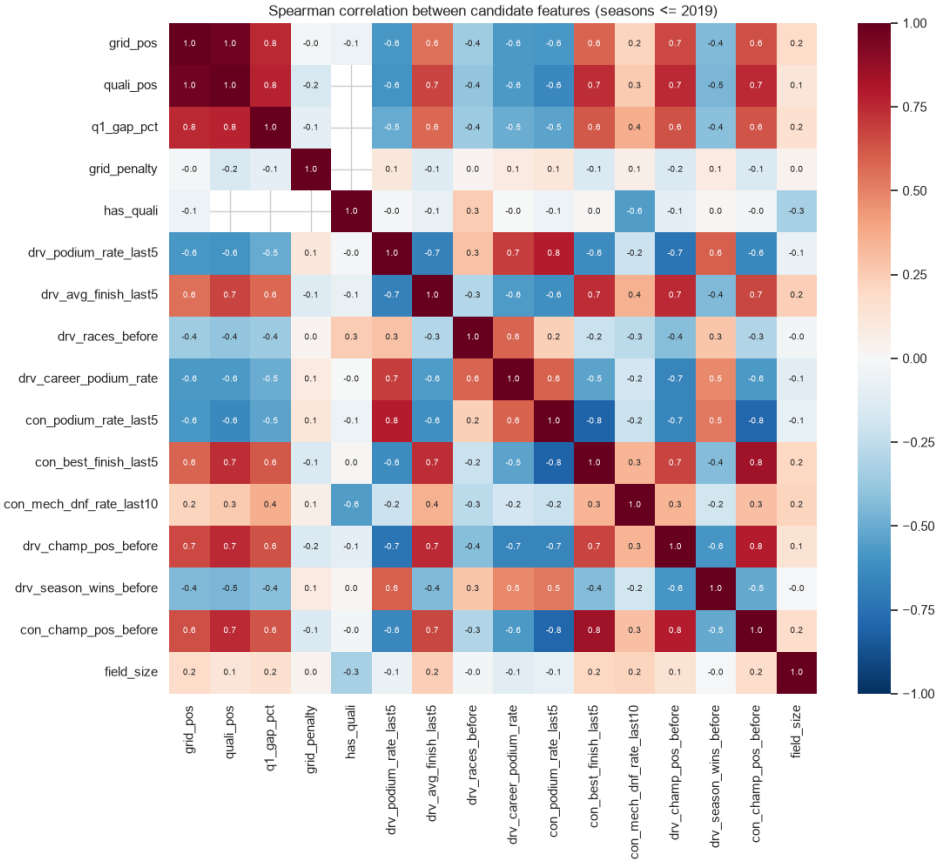


Figure 11. Spearman correlation between the candidate features (seasons up to 2019).

- Grid and qualifying position are almost the same (correlation 0.98, AUC 0.904 vs 0.905 on the same rows), and qualifying position is missing for 63% of rows, so we dropped qualifying position.
- Grid penalty is weak on its own, but drivers who got a penalty podium a lot more often from the same grid band (P7-10: 20.6% vs 6.2%), so we kept it.
- has_quali and field_size are also weak alone. We kept has_quali because it tells the model when qualifying values were filled in, and field_size because it sets the base rate (3 podium places for 20 cars vs 30 cars).

Selected features and why each is known before the race

Group	Feature	Why it is known before the race	AUC
Grid	grid_pos (pit lane = last)	The grid is fixed before the start	0.883
Qualifying	q1_gap_pct (% off the fastest Q1 lap)	Qualifying happens before the race	0.830
Qualifying	grid_penalty (grid minus qualifying slot)	Both are known before the start	0.545
Qualifying	has_quali	Known once qualifying is over	0.507
Driver form	drv_podium_rate_last5	Previous 5 races only	0.812
Driver form	drv_avg_finish_last5	Previous 5 races only	0.795
Driver form	drv_races_before	Count of earlier races	0.661
Driver form	drv_career_podium_rate	Previous races only	0.808
Constructor form	con_podium_rate_last5	Team's previous 5 races	0.829
Constructor form	con_best_finish_last5	Team's previous 5 races	0.815
Constructor form	con_mech_dnf_rate_last10	Team's previous 10 races	0.621
Standings	drv_champ_pos_before	Standings after the previous race	0.837
Standings	drv_season_wins_before	Standings after the previous race	0.702
Standings	con_champ_pos_before	Standings after the previous race	0.829
Context	field_size	Known when the grid is set	0.546

Qualifying features are missing for about 58% of rows (old seasons) and championship positions for 11.8% (first race of a season). These are filled in inside the model pipeline.

6. Train / validation / test split

We split by season instead of randomly because the goal is to predict future races from past ones. A random split would put drivers from the same race in both train and test and make the scores look better than they really are. Everything (models, settings, training years, threshold) was chosen on validation, and we only used the test set once at the end.

Split	Seasons	Races	Rows	Podium rate	Rows with qualifying
Train	1950-2019	1,007	22,444	13.5%	37%
Validation	2020-2021	39	775	15.1%	100%
Test	2022-2024	68	1,352	15.1%	100%

We also checked which years to train on by training the baseline from four different start years:

Train from	Train rows	Val ROC-AUC	Val PR-AUC
1950	22,444	0.929	0.730
1980	15,656	0.929	0.731
1994	10,091	0.930	0.730
2003	6,782	0.929	0.739

The results are very close, so we went with 2003-2019. It scored the best, qualifying data is complete from 2003, and those cars are closer to the ones we are predicting. One side effect is that has_quali is almost always 1 after 2003.

7. Predictive models

PR-AUC is our main metric because podiums are rare (about 15%) and it focuses on ranking the real podium finishers at the top. F1 is our second metric because it looks at the actual yes/no predictions, using the threshold that gives the best F1 on validation. We also report ROC-AUC, precision and recall. All models use the same pipeline (median imputation, scaling, model), fitted on the training rows only.

Model	How it works	Limitations
Logistic regression (baseline)	A weighted sum of the features turned into a probability.	Can't learn interactions unless we add them ourselves.
Random forest	Many decision trees trained on random samples, then averaged.	Slower, harder to explain, and it overfits on our data.
Gradient boosting	Trees added one by one, each fixing the errors of the previous ones.	More settings to tune, overfits if the trees are too deep.
Neural network (MLP, 1-2 hidden layers)	Learns non-linear combinations of the features.	Needs scaling, sensitive to its settings, hardest to explain.

Each attempt is in its own notebook cell and compared to the baseline on validation:

Attempt	Val ROC-AUC	Val PR-AUC	Val F1
Logistic regression (baseline)	0.929	0.739	0.699
Random forest (depth 6)	0.928	0.709	0.714
Random forest (depth 12)	0.929	0.714	0.707
Gradient boosting (slow, shallow)	0.928	0.707	0.711
Gradient boosting (fast, deep)	0.918	0.691	0.698
Neural net (1 layer: 32)	0.907	0.697	0.715
Neural net (2 layers: 64-32)	0.904	0.691	0.697

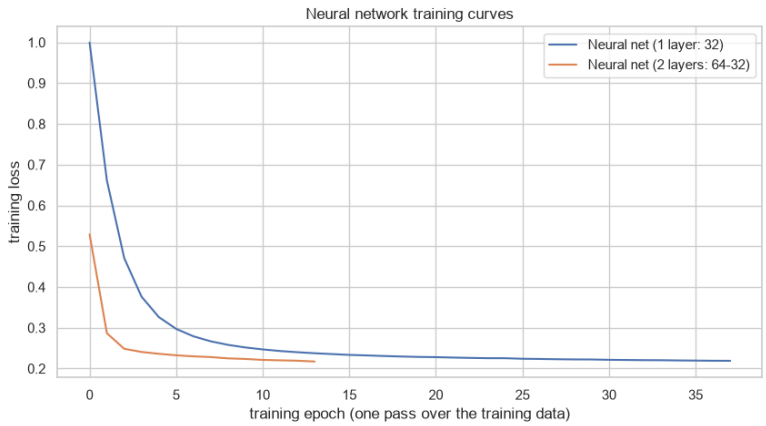


Figure 12. Training loss of the two neural networks.

For each model type we kept the best setting and picked its threshold on validation (LR 0.28, RF 0.29, GB 0.25, NN 0.50). This is the only time we used the test set:

Model	Data	ROC-AUC	PR-AUC	F1	Precision	Recall
Logistic regression	Train	0.929	0.685	0.666	0.575	0.792
	Validation	0.929	0.739	0.699	0.659	0.744
	Test	0.936	0.708	0.685	0.591	0.814
Random forest (depth 12)	Train	0.979	0.899	0.778	0.679	0.912
	Validation	0.929	0.714	0.707	0.667	0.752
	Test	0.941	0.710	0.684	0.588	0.819
Gradient boosting (slow, shallow)	Train	0.959	0.801	0.716	0.608	0.871
	Validation	0.928	0.707	0.711	0.662	0.769
	Test	0.939	0.721	0.688	0.604	0.799
Neural net (1 layer: 32)	Train	0.928	0.679	0.631	0.690	0.580
	Validation	0.907	0.697	0.715	0.760	0.675
	Test	0.921	0.660	0.556	0.651	0.485

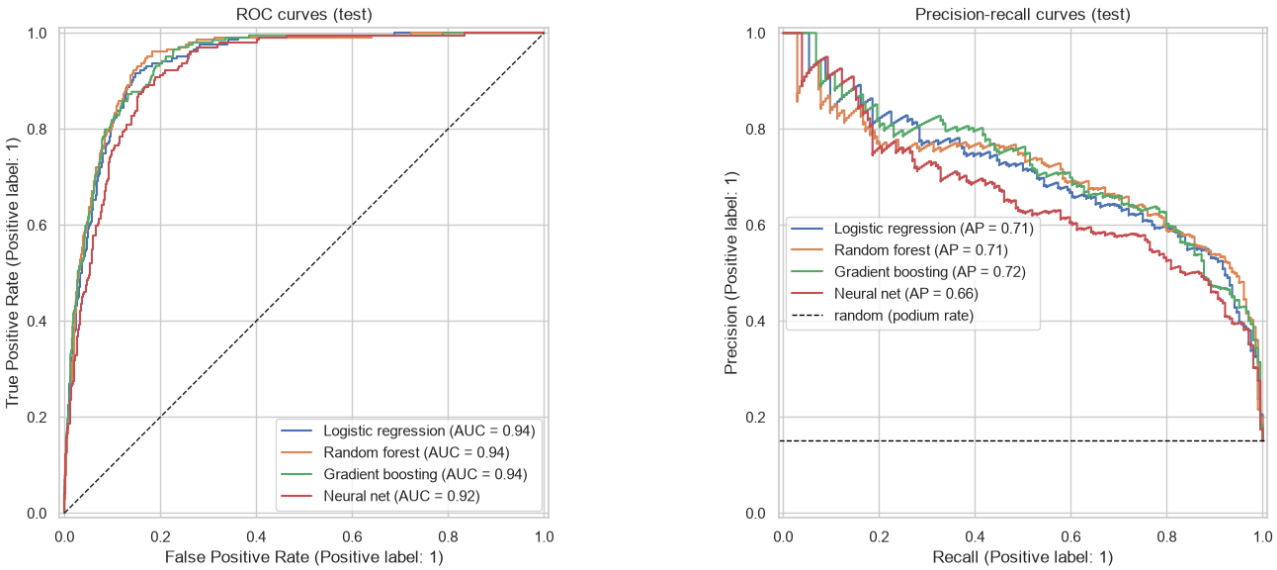


Figure 13. ROC and precision-recall curves of the four final models on the test set (dashed line = random).

- All models do much better than random (test PR-AUC 0.66-0.72 compared to 0.15).
- Gradient boosting has the best test PR-AUC (0.721), but random forest (0.710) and the baseline (0.708) are very close, and the baseline was the best on validation. So the more complex models don't really pay off here.
- Random forest and gradient boosting overfit (train PR-AUC 0.90 and 0.80 vs about 0.71 on test). The baseline stays about the same on all three sets.
- The neural network is the weakest. Its threshold of 0.50 turned out to be too strict for 2022-2024, so it only found 49% of the podiums (F1 0.56).
- We picked logistic regression as the final model because it was the best on validation.

8. Experiments

All experiments use the final logistic regression. Preprocessing fills every missing value in the training data (e.g. 483 missing championship positions) and scales every feature to mean 0 and standard deviation 1. Before that the ranges were very different (races of experience go up to several hundred while the rates are between 0 and 1).

Feature-group ablation

We removed one feature group at a time, retrained the model and looked at how much validation PR-AUC dropped.

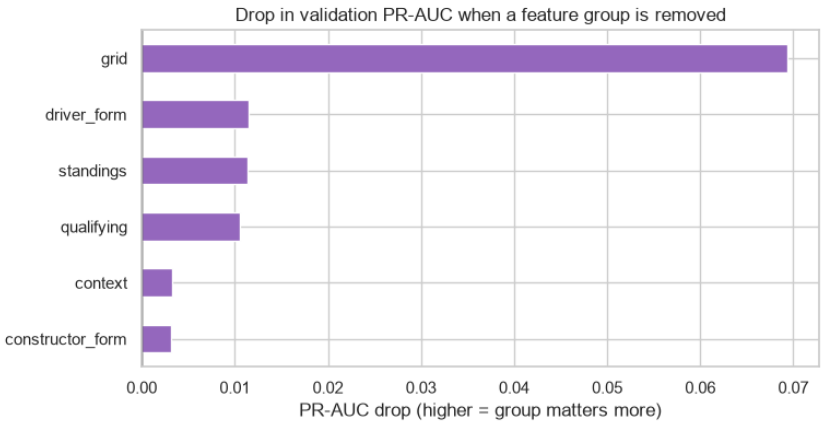


Figure 14. Drop in validation PR-AUC when one feature group is removed.

Grid matters by far the most: removing it drops PR-AUC by 0.069 (from 0.739 to 0.669). Qualifying, driver form and standings each cost about 0.01, and constructor form and context about 0.003. The drops are small

because the groups overlap, a fast car shows up in team form, driver form and standings at the same time. No group made the model worse, so we kept all of them.

Preprocessing-order experiments

A: imputation and scaling fitted on all the data before the split (leaky) instead of on the training rows only. B: scaling first, then imputation.

Setup	Val ROC-AUC	Val PR-AUC
Impute then scale, fitted on train only (ours)	0.9292	0.7386
A: impute then scale, fitted on all data before the split	0.9292	0.7368
B: scale then impute, fitted on train only	0.9291	0.7386

In A the medians change (e.g. q1_gap_pct goes from 1.342 to 1.223), so information from validation and test does get into training. The score only changes in the third decimal because there are very few missing values after 2003, but fitting on the training data only is still the correct way. In B the results are basically the same (less than 0.001 difference), so we kept the usual order.

9. Explainability (XAI)

We explain the final model on the test set. **Note:** these results show what the model relies on, not what causes podiums. For example, grid position matters because it reflects how fast the car is, not because the starting slot itself wins races.

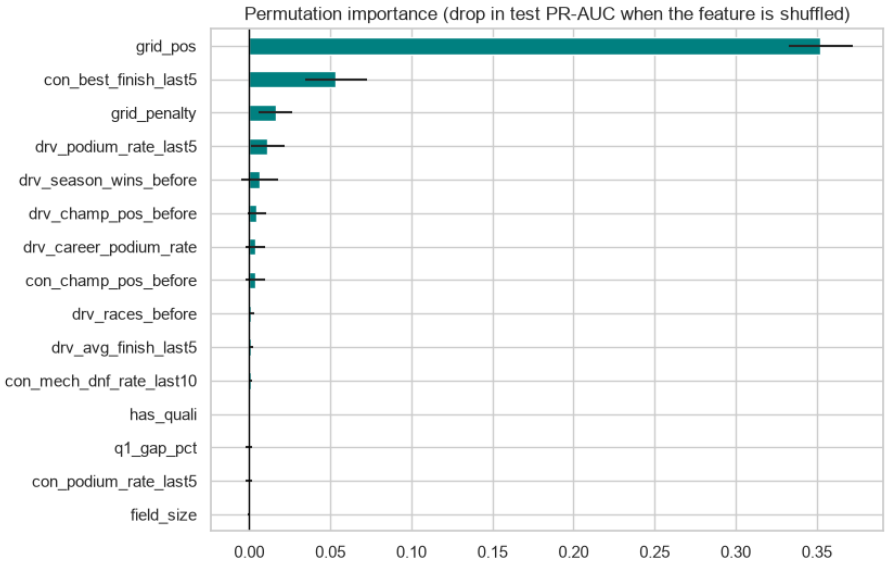


Figure 15. Permutation importance on the test set (drop in PR-AUC when a feature is shuffled, 10 repeats).

Grid position is by far the most important (about 0.35 drop), then the team's recent best finish (about 0.05), the grid penalty and the driver's recent podium rate. Most other features are close to 0 because they overlap and cover for each other.

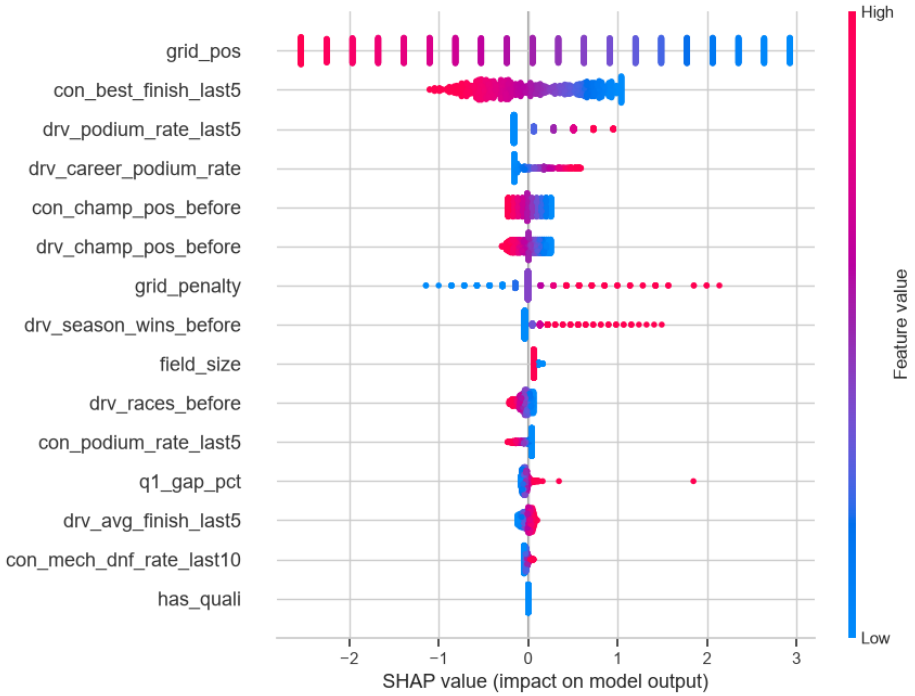


Figure 16. SHAP beeswarm on the test set (red = high feature value, blue = low).

- Starting near the front pushes strongly towards a podium. Good recent team results, driver form and wins this season also push it up.
- A big grid penalty pushes it up too, since a driver who qualified much higher is faster than his grid slot shows.
- A few small effects go the other way (e.g. team podium rate). These features are strongly correlated and the model balances them against each other, so they should be read together.

Local explanation: one prediction

For the local example we picked the 2022-2024 podium finisher with the biggest grid penalty: Max Verstappen at the 2022 Belgian GP. He qualified on pole, got a 13 place engine penalty, started 14th and won. Our model gave him 57%, while only 1.6% of drivers starting 14th made the podium in 2003-2019.

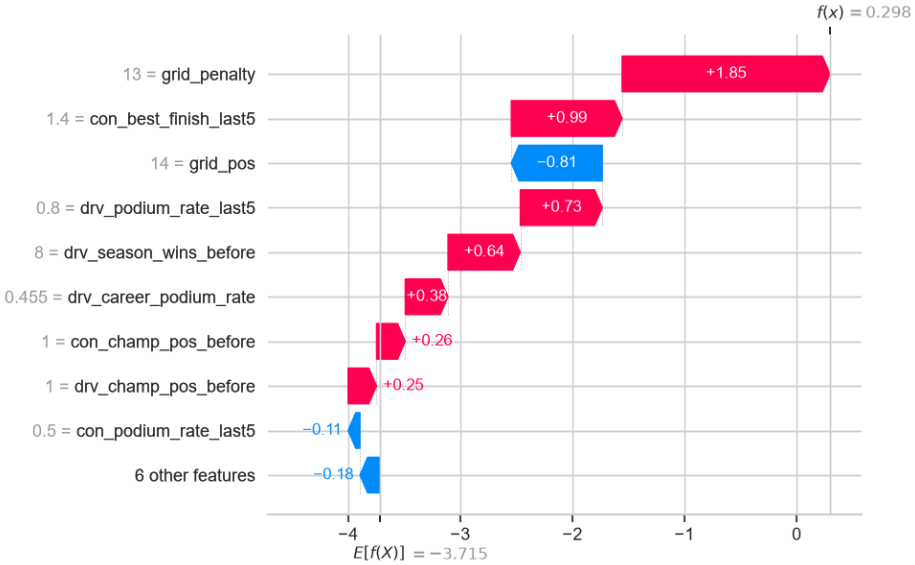


Figure 17. SHAP waterfall for Verstappen, Belgian GP 2022 (log-odds, f(x) = 0.30 means 57%).

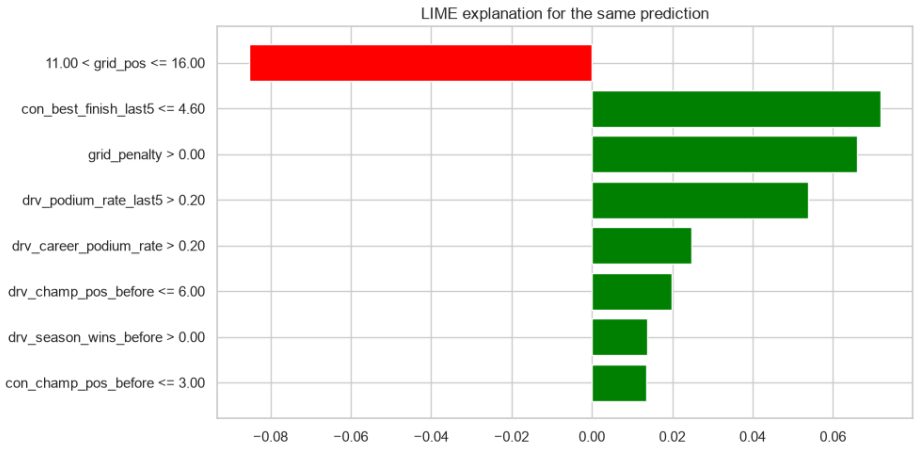


Figure 18. LIME explanation for the same prediction.

SHAP shows that the grid penalty (+1.85), the team's recent results (+0.99), his recent podium rate (+0.73) and his 8 wins so far (+0.64) outweigh the bad grid slot (-0.81). LIME agrees, the grid slot is the only big negative. This is exactly the kind of case we kept grid_penalty for.

10. Inference function

`predict_podium(record)` takes a driver's pre-race features as a dictionary and returns the podium probability, a yes/no prediction (threshold 0.28) and a list of the features it had to fill in. Missing features are filled with the training medians through the same pipeline.

Record	Missing features	Probability	Prediction
Complete: Lando Norris, 2024 Abu Dhabi GP (pole, 2nd in the championship, he won)	None	62.9%	Podium
Partial: same driver without q1_gap_pct, grid_penalty, has_quali and both championship positions	5	51.9%	Podium

With missing inputs the probability moves towards an average driver (for example championship position 11 instead of 2), but pole position and recent form still keep it as a podium.

11. Known limitations

- **Qualifying coverage drift:** there is no qualifying data before 1994 and it is patchy until 2002, so we train from 2003 and lose the older data.
- **Era sensitivity:** new regulations (2014, 2022) change which teams are fast, so the first races of a new era are harder to predict.
- **Small evaluation sets:** validation has 39 races and test 68, so differences of about 0.01 between models are within noise.
- **Overlapping features:** form and standings are correlated, so their importances should be read as a group.
- **Missing inputs:** missing features are filled with medians, which pulls predictions towards the average.
- **Team renames:** a renamed team gets a new constructorId, so its form starts from zero.
- **Not causal:** the explanations describe what the model uses, not what causes podiums.